# Web Technology
## UNIT-1

## Introduction:

* History of web and internet:

- In 1960s, the concept of packet switching emerges as a way to transmit data efficiently.
- Later in 1970s. ARPANET became the first network to use packet switching it was established to connect computers at research institutions.
- Further in 1980s, TCP/IP protocol suite became the common language for various networks to connect or exchange data.

* Web development strategies:

(i) Responsive web designing: (Screen size adaptibility)
(ii) Performance optimization: (fast loading and better ranking)
(iii) UCD(user centered design): (focus on user's area)
(iv) Security measures: prevent from data breaches, keeps Software updated and use robust security protocols)
(v) Cross-browser compatibility: Support multiple browser
(vi) CSS frameworks: UI enhancement
(vii) Content management system (CMS): allows simple content editing for non-technical users.
(viii) Meta information or SEO (search engine optimized): additiona data to identify clients computer)
(ix) Accessibility: accessible to everyone
(x) Automated testing: ensures code quality and catches bugs early in development process

* Network governing protocols:

↳ Directly governing the web:

- HTTP (Hypertext transfer Protocol): It defines how web brows and servers communicate, how a browser requests a web page from a server.

• HTTPS (Hypertext transfer Protocol Secure): It encrypts the communication b/w the browser and the server protecting sensitive data like login credentials, card information, etc.

↳ Indirectly supporting the web:

• TCP/IP (Transmission control Protocol / Internet Protocol): TCP is responsible for transmission of data into packets and ensures reliable delivery.
IP assigns unique address to device for network routing

• DNS (Domain name System): It translate human readable domain names into machine readable IP addresses that computers use to locate web servers.

↳ Some additional protocols that don't involved in web browsing but are crucial for maintaining overall ecosystem

• FTP: File transfer protocol used for transferring files b/w computers.

• Email: uses separate protocols for sending (SMTP) and to receive.

(POP3 / IMAP) → Internet message Access Protocol (keeps e-mails on the server and allows access from multiple device)
↓
(Post office protocol) (retreive emails from a mail server to user)

• Telnet: An older protocol for remote access. It allows you to log into another system and browse files and dictionaries on that remote system.

• UDP: User datagram protocol is a connectionless protocol for data transfer it is faster but less reliable.

ICMP (Internet control message Protocol): It does not directly involved in data transmission but carries information-al message b/w network devices, like error messages indicating unsuccessful packet delivery.

## Writing Web Projects:

- Write a project mission statement
Write the specific mission statement that we want to do for what purpose the particular website is going to be created.

- Identify the objectives:
  - S → specific
  - M → measurable
  - A → atainable
  - R → realistic
  - T → time limited

- Identify your target users: It can be done by
  - market research
  - focus groups
  - understanding audience needs.

- determine the scope: List all the features and functionalities of your websites and priartize them based on importance and complexity.

- Budgets:
  - Assumed budget
  - Additional cost
  - Final budget

- Planning Issues: Plan how the content will be accessed on your website.

- Technical requirements: What technologies are going to be used.

- Development:
  - Front-end
  - Back-end
  - content

- Testing and maintainance:
  - Test the code
  - maintain the web server
  - fix the bugs
  - Provide regular updates
  - Take user's feedback

- Register with ISP and Launch:
- Register with domain name
- Get space on web
- connect domain name with server
- Host the website

## * Internet services and tools:

- Internet services allows us to access huge amount of data such as text, graphics, sound, softwares, etc. over the intern
- Internet services have mainly four categories as follows:

1) communication services
2) Information retreival services
3) Web services
4) World wide web

(i) communication services: They allow us to communicate or exchange information over internet.

eg → E-mail
↳ Instant message
↳ mailing List
↳ IRC (internet relay chat)

ii) Information retreival services: Allows easy access to information over internet.

eg → FTP (use to transfer file)
Archie (helps to search file by name)
Gophor (display documents on remote sites)

(iii) Web service: Allows exchange of information b/w applica on the web

(iv) WWW: It offers a way to access information spread ov several server on the internet.

eg → Video conferencing
↳ information like that text, audio, video,
↳ hyperlink etc can be accessed easily.

* client server computing: It is fundamental architectur distributing tasks and resources accross a network.

↳ It relies on two key players, client and server.

i) client: These are software program or devices that ini requests for resources or service.

↳ They have a user interface to interact with the server.

eg → Mobile apps, email clients, etc.

ii) Server: These are powerful computers or program th fullfill client requests.

↳ They store data, perform calculations, and manage r resources. They are capable of handle multiple clien simultaneously.

## Working of client-server computing:

Request: client sends request to the server
↳ Request specifies desired resources or service, such as web page, a file download or a database query.

- Processing: Server receives the request process it and perform the requested operations.
- Response: Server sends response back to the client
↳ Response may be a confirmation message or an error message.

## * Advantages of client-server computing:

- easy backup and update
- additional servers can be added to ease the workload
- easy to share resources
- Centralized storage allows more secure access to data and application

## * disadvantages of client-Server computing:

- If server fails, all connected clients are affected and loose access to resources.
- Initial cost is high to setup a client-server computing resources.
- Server can go down in case of no. of request of client exceeds the server's processing capacity.

## * Types of client-server models:

1) one-tier archiecture
2) Two- " "
3) Three - " "
4) N-tier "

## * Examples.

- Web-browser (chrome, firefox)
- Email client (G-mail)
- Social media Platforms (facebook, Twitter)

HTML: HTML stands for Hyper text markup language. It is the basic building block of the web, and it's used to create and structure web pages and their content.

## Why HTML is important:

Structure: HTML is what gives structures to a webpage, organizing headings, Paragraph, images.

- Easy to Learn: HTML is simple and user-friendly
- Universal Language: every websites you visit is built using HTML

* HTML Tags: are like Keywords which define that how web browser will form at and display the content

* Heading Tag:

<h1> - - - . </h1>

:

<h6> - - - . </h6>

- <br> Tag: br stands for break line
- <img>: insert image
- <li>: items in list
- <p> : Paragraph
- <table>: create table
- <i>: Italic text
- <b>: bold text

* HTML Table: consists of table cells inside rows and columns

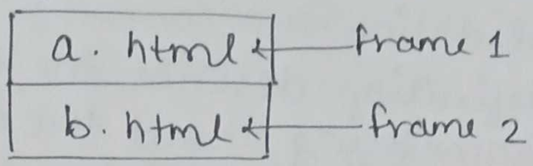
Ex -
```
<table>
  <tr> <th> Company </th>
       <th> contact </th>
       <th> country </th>
</tr>
  <tr>
   <td> XYZ </td>
   <td> XYZ </td>
   <td> Germany </td>
</tr>
  <tr> <td> XYZ </td>
       <td> XYZ </td>
       <td> USA </td>
</tr>
</table>
```

td: Table data
tr: Table Row
th: header cell in a table

frames: It can display one or more than one html document in the same browser window

• Each html document is called FRAME and each frame is independent of other.

• <frameset> tag is used divide browser




Attribute of frameset Tag:

• Row
• column
• None
• Border color
• Frame border

forms in HTML:

An HTML form is used to collect user input. The user input is most often sent to a server for Processing.

Tag:- <form>
</form>

<label for = "fname"> first name: </label>
<input type = "text" id = "fname"

* List: An HTML list is a record of related information used to display the data or any information on webpages in ordered or unordered form.

Types of List:

1) unordered List: An unordered list starts with <Ul> tag. each list items <li> tag

eg- <Ul>
<li> coffee </li>
<li> Tea </li>

2) ordered List- An ordered list starts with <ol> tag. each list items starts with the <li> tag

eg - <ol>
<li> coffee </li>
<li> Tea </li>
</ol>

* XML DTD: XML (Extensible markup language)
* XML Document type definition.
* used to describe XML language precisely.
* used to define structure of a XML document.
* contain list of legal elements
* used to perform validation
* A document type definition describe the three structure of a document and something about the data

* Purpose of DTD: Its main purpose is to define the structure of an XML document
↳ contain a list of legal elements and define the structure with their help of them

* Types of DTD → 1. Internal DTD 2. external DTD

1) Internal DTD: defined within the XML document itself, by using the <!DOCTYPE> declaration
↳ DTD is contained within the angle bracket of the <!DOCTYPE> declaration, and is placed before the root element of the document.

SYNTAX (Internal DTD)

```
<!DOCTYPE> root_element [
<!Element> root_element + (child_element) >
  ]>
    <root_element>
    <child_element> Some Text </child>
    </root_element>
```

external DTD: An external DTD is defined in a seprate file and its referenced by the XML document using the system keyword in the <!DOCTYPE>.

SYNTAX: 

```
<!DOCTYPE root_element SYSTEM "example.dtd">
    <root_element>
    <child_element> Same Text </child_element>
    </root_element>
```

XML Schemas: commonly known as XML Schema definition XSD. It is used to describe and validate the structure and content of XML data.

↳ It is like DTD but provides more control an XML structure

Syntax

<xs: schema xmins: xs=" ">

* Definition Type

1) Simple Type used only in the context of the text
   Ex → xs: int, xs: string

2) complex Type - It is the container for other element, definition allows you to specifiy which child element an element can contain to provide some structure within your XML document

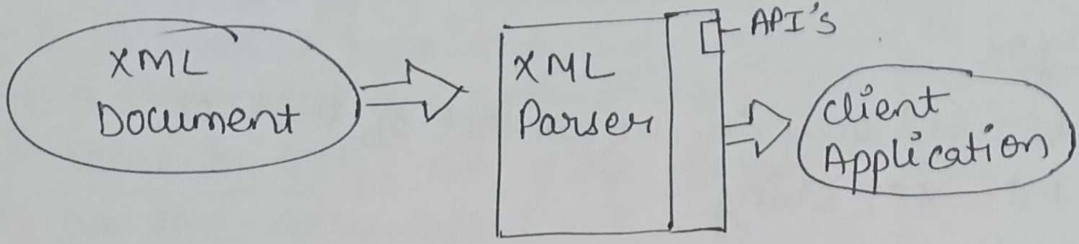
* Application of XML:

- XML is a markup langⁿ based on standard generatized markup langⁿ used for defining markup language.
- It is used to encode information for documentation.
- It is used to transfer data b/w two system

* Difference b/w DTD & XSD:

| DTD | XSD |
|---|---|
| • Document type definition | XML schema definition. |
| • DTD are derived from SGML syntax | XSD are written in XML. |
| • DTD doesn't support data type | XSD supports data type for elemen and attribute. |
| • DTD doesn't support namespace | XSD support namespace |
| • DTD is not extesible | XSD exteasible |
| • DTD is not support to learn | XSD is simple to learn |

* **XML Parsers**: software library (or Package) that Provides method's (or interfaces) for client application to work with XML document
↳ XML parser is designed to read the XML and create a way for programs to use XML



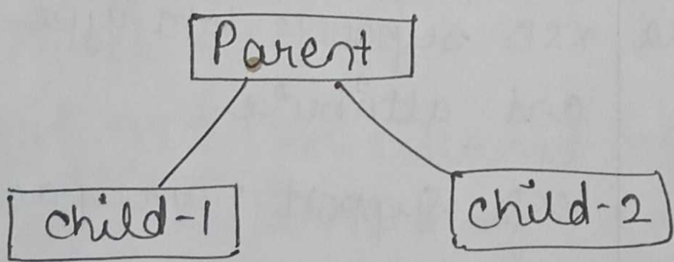

* Types of XML Parsers:
 1) DOM
 2) SAX

* DOM: Document object model
 ↳ object which contains all the information of an XML document
↳ It is composed like tree structure
↳ implements a DOM API
↳ API is very simple to use

Advantages -
It support both read and write operations and the API is very simple to use.

Disadvantages - It is comparatively slower than other parsers.

ex -



SAX: SAX Parser implement SAX API. This API is an event based API

does not create any internal structure.

, clients does not know what method to call, they just override the methods of the API.

Advantages:

- It is simple and memory efficient.
- It is very fast and work for huge document

Disadvantage:

- client never know the full information because the data is broken into pieces.

* diff b/w HTML & XML

| HTML | XML |
|---|---|
| • Hypertext markup langⁿ | extensible markup langⁿ |
| • Static in nature | Dynamic |
| • Ignore small errors | does not ignore |

* diff b/w <Div> and </span> tag

| Div | Span |
|---|---|
| • Block level element | Inline |
| • used to wrap section of a document | used to wrap small portions of text, image. |
| • used while create CSS base layout in HTML | used to stylize text. |